

Chapter ∞ . Review for Final Exam

ARIN7013 Numerical methods in artificial intelligence

The University of Hong Kong



- Convolutional Neural networks (CNNs):
 - what is the inductive biases for CNNs, i.e., what are the features of CNNs compared with the fully connected neural networks?
 - What is the cross-correlation?
 - For a given CNN with a certain layer, can you figure out the output from each layer and the number of parameters?
 - Can you use backpropagation for a convolutional layer to calculate the gradient of the loss function with respect to the parameters?
- Kernel methods
 - positive definite kernels, Aronszajn Theorem, kernel trick
 - representer Theorem
 - Kernel Ridge Regression: how to derive the matrix form?

- Direct methods for solving linear equation system:
 - LU decomposition, Gauss elimination method. What are the computational complexity? Given a proper matrix, can you perform LU decomposition for it, and use this decomposition to solve a linear equation system?
 - What is pivoting, and what is it used for?
 - What is the stability of a linear equation system?
 - What's the condition number?
- Iterative methods for solving linear equation system
 - Jacobi Method and Gauss-Seidel method: computational complexity
 - What's the error equation?
 - How do they converge?
- Low-rank matrix approximation: how to get the low-rank approximation, and why do we need it?

- Singular Value Decomposition (SVD)
 - How to get SVD for a given matrix?
- Numerical methods for solving the eigenvalues of a given matrix
 - Power iteration method, inverse iteration method and Rayleigh-quotient iteration method: how to implement them?
 - What are the conditions to make them converge, and what are the convergence rates?
- (Kernel) Principle Component Analysis (PCA):
 - How to use (Kernel) PCA to find the dominant components for a given set of points?
 - Why (kernel) PCA works?

- How to solve ODEs numerically, and what are their stability and convergence rates?
- What is the ResNet?
- What is Neural ODEs, and how to compute the gradient?
- Do you know the memory requirements for ResNet and Neural ODE?

Summary

- This course introduces several learning methods: CNNs, kernel methods (especially kernel ridge regression method), (kernel) PCA, ResNet and neural ODE
- This course also covers several topics related to these learning methods: direct methods and iterative methods for solving linear equation system, low-rank matrix approximation (especially SVD), eigenvalue solvers, ODEs and its solvers
- It seems that you have learned many tools to complete several learning tasks. However, you should be skeptical: Why do you choose one method not the other one?

- Time: 18:30 - 21:00, May 18, 2026.
- Format: 8 long questions. You are advised to write the details steps or derivations.
- There are questions on your understanding of the methods themselves. For these type of questions, you just need to write your own opinion or idea on them. No exact calculations are needed.
- Only approved calculators as announced by the Examinations Secretary can be used in this examination.
- Candidates are permitted to bring to the examination ONE sheet of A4-sized paper with hand-written notes on both sides. If it is NOT hand-written, then it will be taken away.

Thanks for your attendance, and good luck with your study!